



B.TECH PROJECT • MID-SEMESTER EVALUATION

Dept. of Electronics & Communication Engineering, NIT Silchar

Wideband MIMO Antenna with Metasurface

Checkpoint 1 — a CPW-fed wideband antenna with a split-ring metasurface reflector

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Under the guidance of

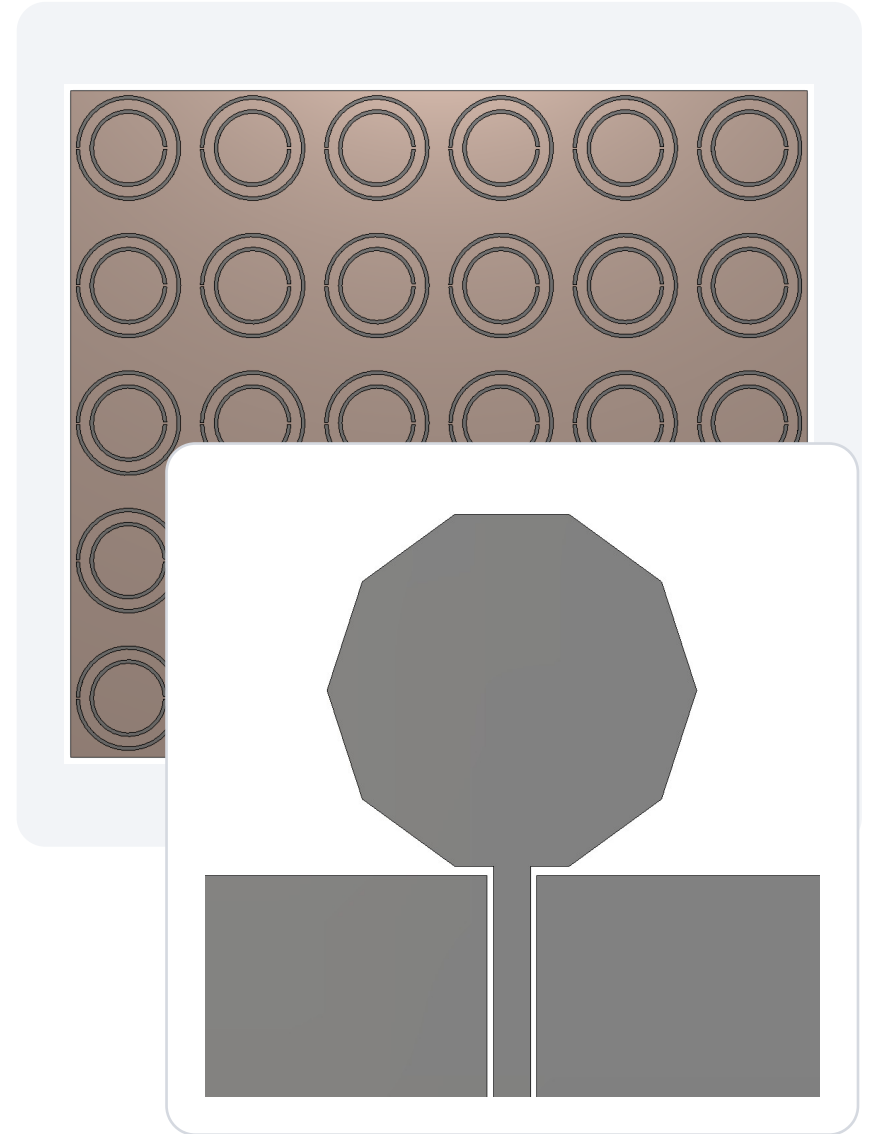
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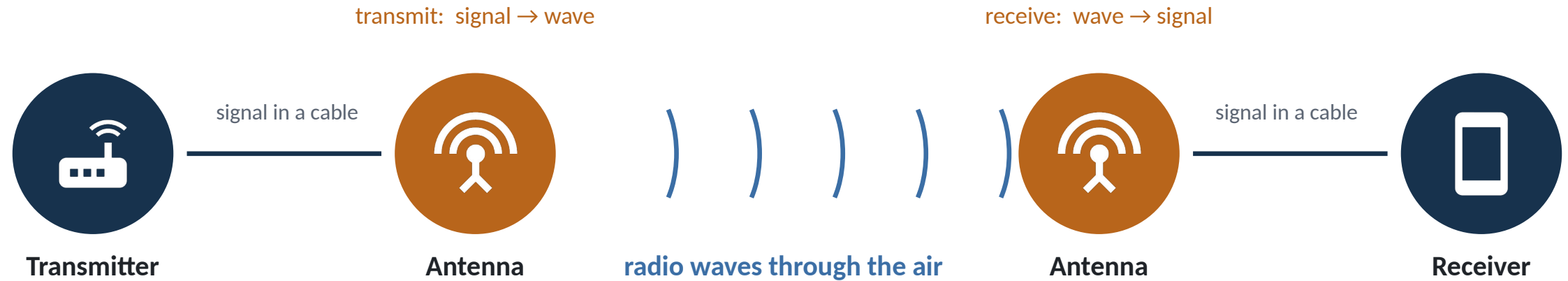
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our antenna (front) + split-ring metasurface (behind)

What is an antenna?



A converter

Electrical signal $\leftrightarrow$ radio wave, in both directions



Everywhere

Phones, Wi-Fi routers, GPS, radar



Size follows wavelength

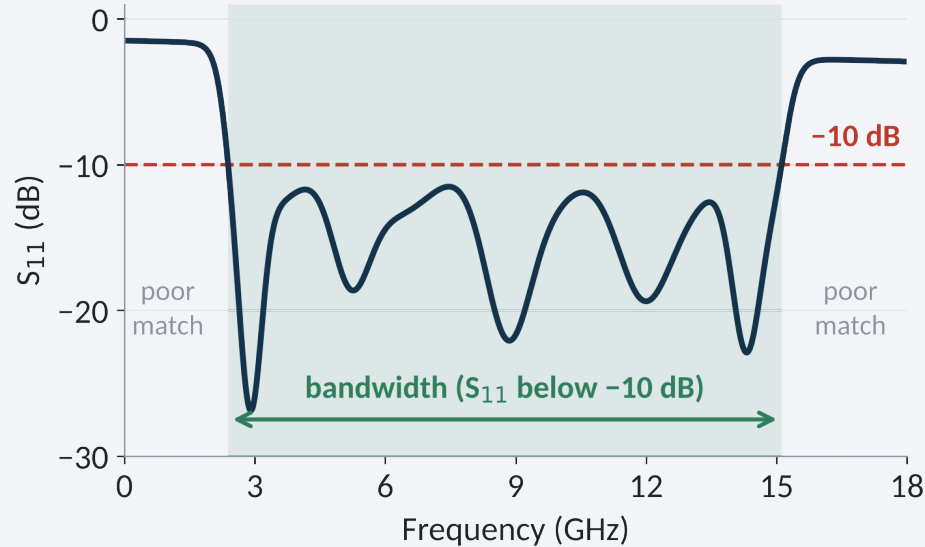
Higher frequency $\rightarrow$ shorter wave
 $\rightarrow$ smaller antenna

How do we judge an antenna?



Matching — S_{11}

how much power bounces back

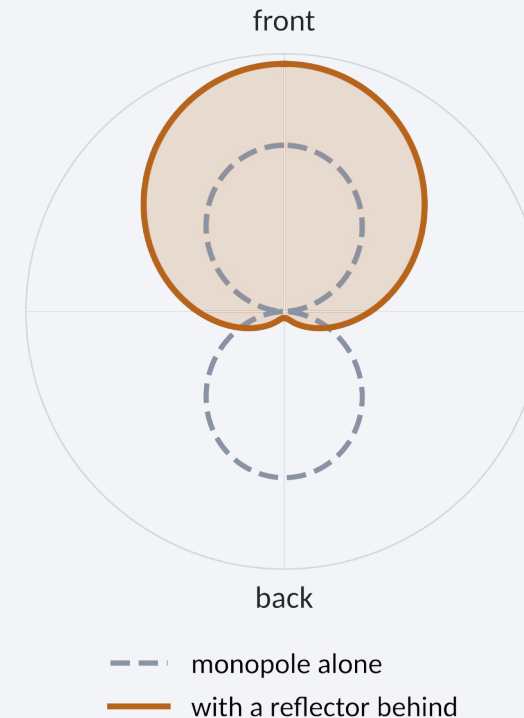


Below -10 dB $\rightarrow$ less than 10 % reflected $\rightarrow$ good match



Gain & radiation pattern

how strongly, and in which direction



Gain (dBi): power in the best direction, compared with an antenna that radiates equally everywhere

A monopole sends **front and back** $\rightarrow$ part of the power is wasted

A reflector behind it sends more power **forward** $\rightarrow$ higher gain

concept sketches, not our results

Why wideband, why MIMO — and what is the problem?



Wideband

One antenna covers many services
(about 2–15 GHz)



MIMO

Several antennas on one device →
more data, more reliable link



Printed monopole

Small, cheap, easy to make and
naturally wideband



The problem

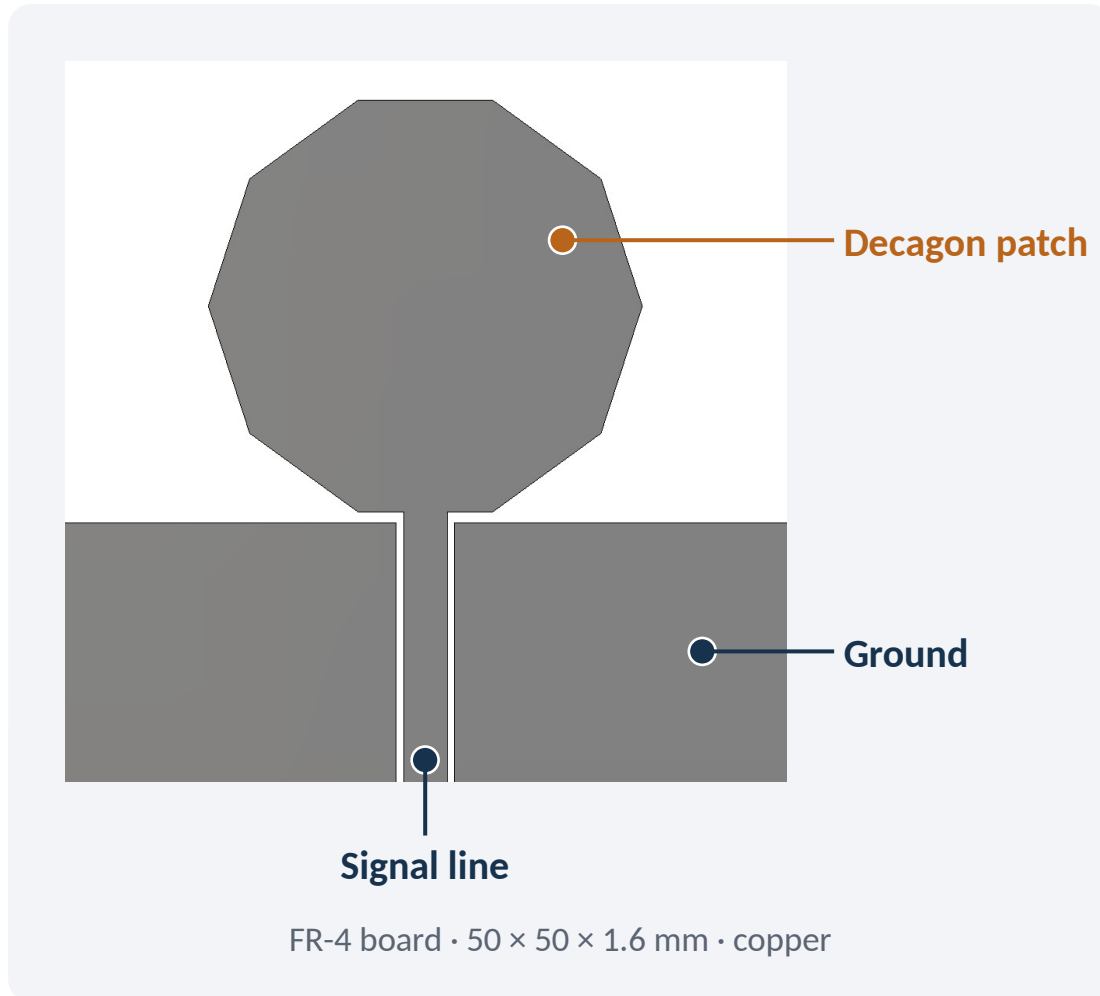
- A printed monopole radiates **to the front and the back** → part of the power goes the wrong way → **low gain**
- A plain metal sheet behind it must sit **about a quarter-wavelength away** → the antenna becomes **thick**



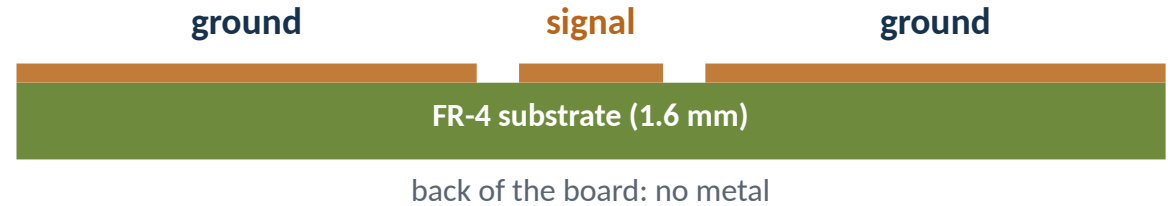
Our question

How can we raise the gain and still
keep the antenna thin and
wideband?

Our antenna: a CPW-fed printed monopole



CPW feed — cross-section



Why printed on FR-4?

Cheap, light and easy to fabricate



Why a decagon patch?

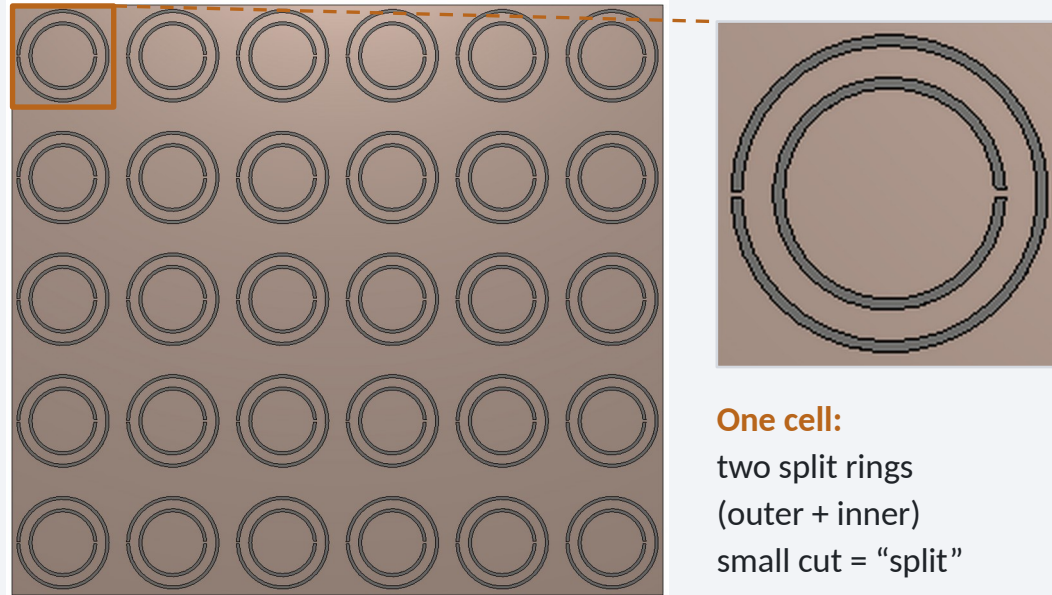
Close to a circle → smooth current paths → several resonances merge into one wide band



Why a CPW feed?

Signal and ground on one side → one copper layer, no vias, easy connector, wideband

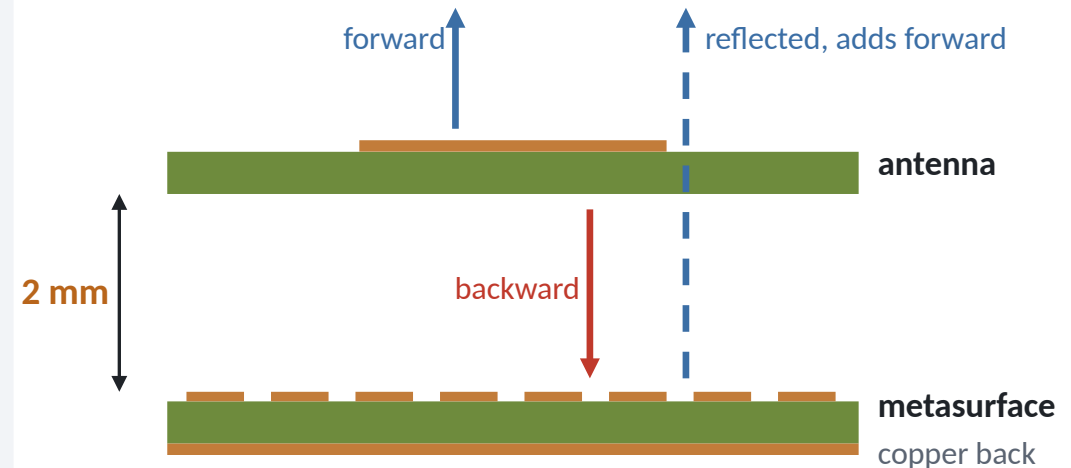
Our idea: a metasurface reflector



6 × 5 cells (top view)

One cell:
two split rings
(outer + inner)
small cut = "split"

Side view (not to scale)



Thin board with repeated small metal cells



Split ring = tiny resonator → two rings, two resonances



Placed only 2 mm behind → can stay thin

Goal: send the backward wave forward → **more gain**
CST: peak directivity **3.7 → 8.8 dBi** (slide 14)

Four objectives

1



Design

A compact CPW-fed wideband antenna covering about 2–15 GHz

2



Optimise

Ground position, then patch size, in CST, for S_{11} below -10 dB

3



Metasurface

Split rings behind the antenna to raise the gain while keeping it thin

4



Extend & test

Phase II: MIMO version, fabrication and measurement

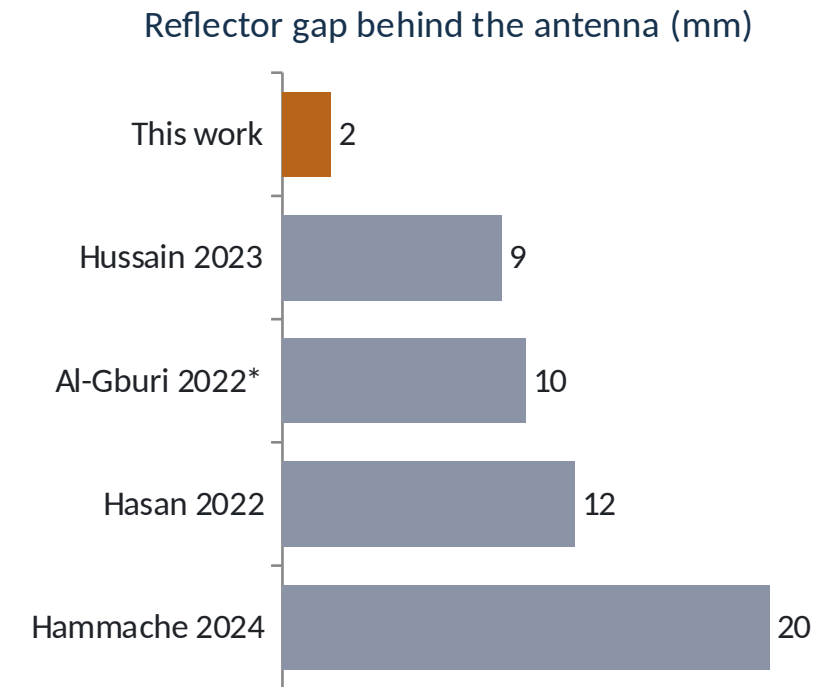
Objectives 1–3: this phase • Objective 4: Phase II

Reflectors add gain, but sit 9–20 mm away

Work	Reflector behind the antenna	Gap	Peak gain
Al-Gburi 2022 [7]	19 × 19 loop FSS	10 mm*	6.7 → 11.5 dBi
Hussain 2023 [8]	5 × 5 ring-frame FSS	9 mm	6.5 → 10.5 dBi
Hammache 2024 [9]	7 × 7 FSS	20 mm	2.2 → 8.4 dBi
Sen 2017 [10]	double split-ring metasurface	—	≈ +5.5 dB
Hasan 2022 [11]	10 × 10 split-ring MS (MIMO)	12 mm	5.4 → 8.3 dBi
This work	6 × 5 double split-ring MS	2 mm	3.7 → 8.8 dBi†

FSS = frequency selective surface · MS = metasurface · * total height · — not given in the abstract

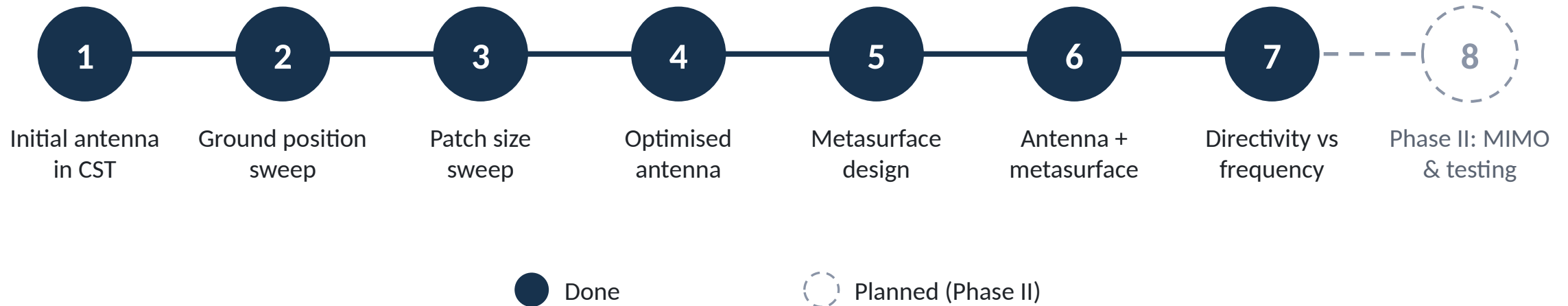
† directivity, $\phi = 0^\circ$ plane (CST)



Research gap

A metasurface only a few mm behind an antenna that works from about 2 to 15 GHz is rarely reported → our aim: 2 mm, then MIMO

Eight steps — seven done



Simulation tool

CST Studio Suite 2019, frequency-domain solver, 0–18 GHz



Material

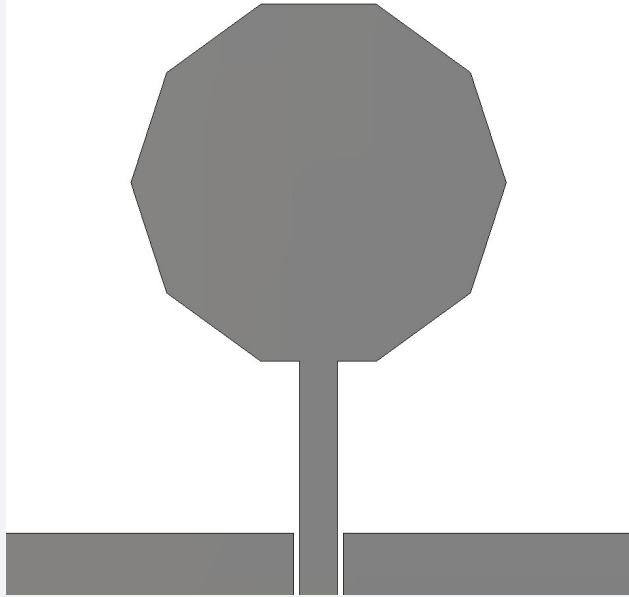
FR-4 board (1.6 mm) with copper conductors



Method

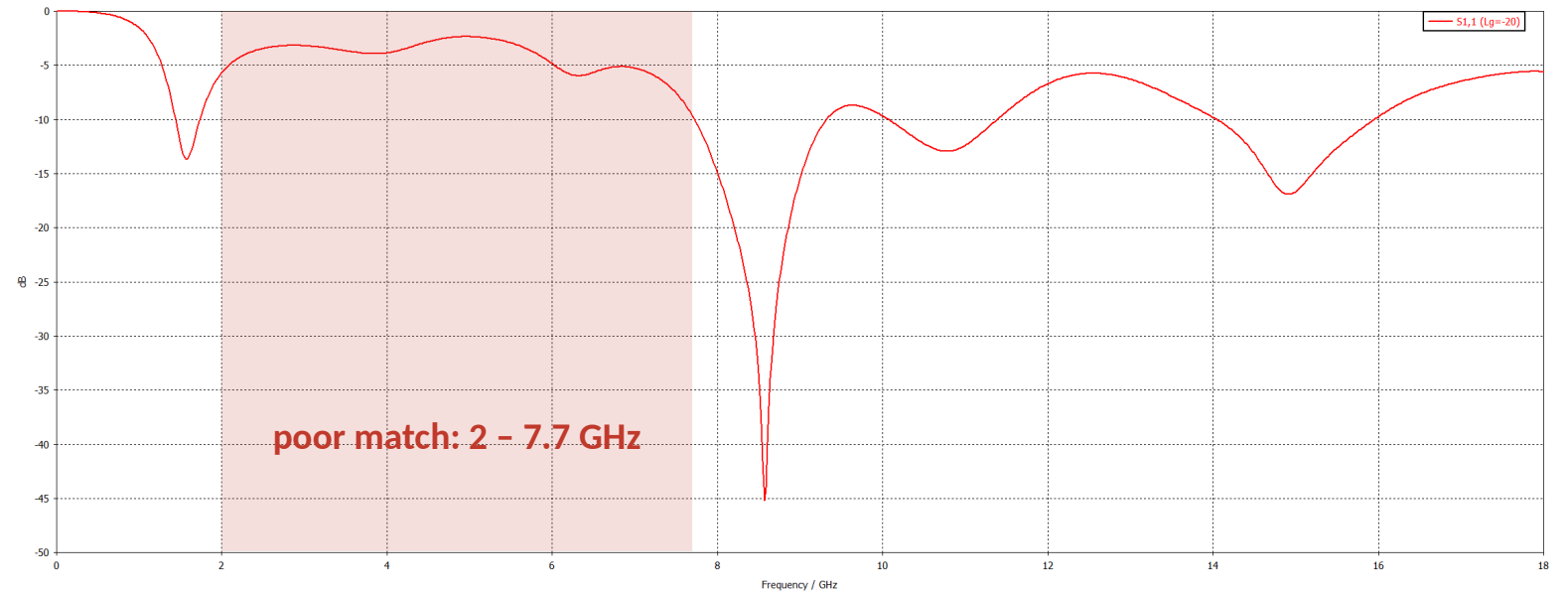
Change one dimension at a time; keep the value with the best S_{11}

Starting point: the initial antenna is poorly matched



Initial design: ground far from the patch
($L_g = -20$ mm)

S_{11} of the initial antenna

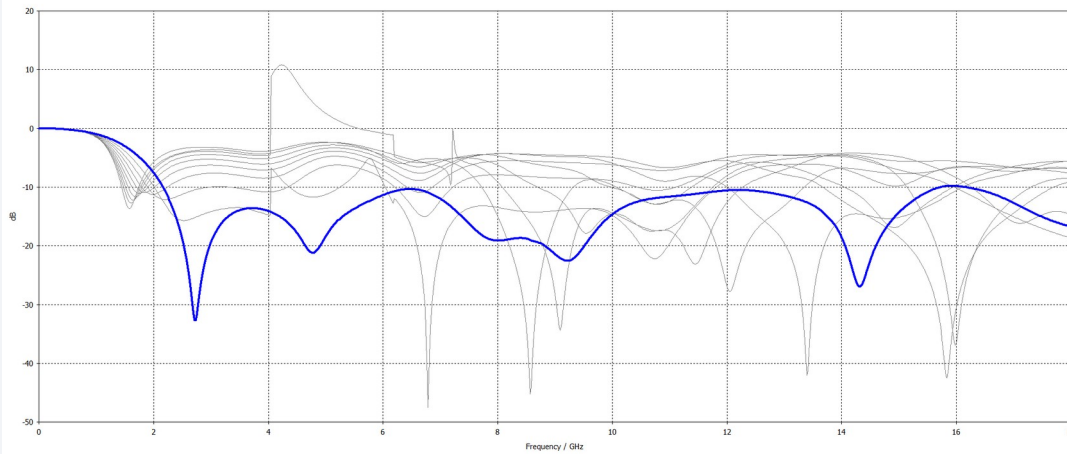


S_{11} stays above -10 dB from 2 to 7.7 GHz; only a few narrow dips below it higher up → not usable yet

Optimisation: ground position first, then patch size



Step 1 • Ground position



— Best: $L_g = -7$ mm

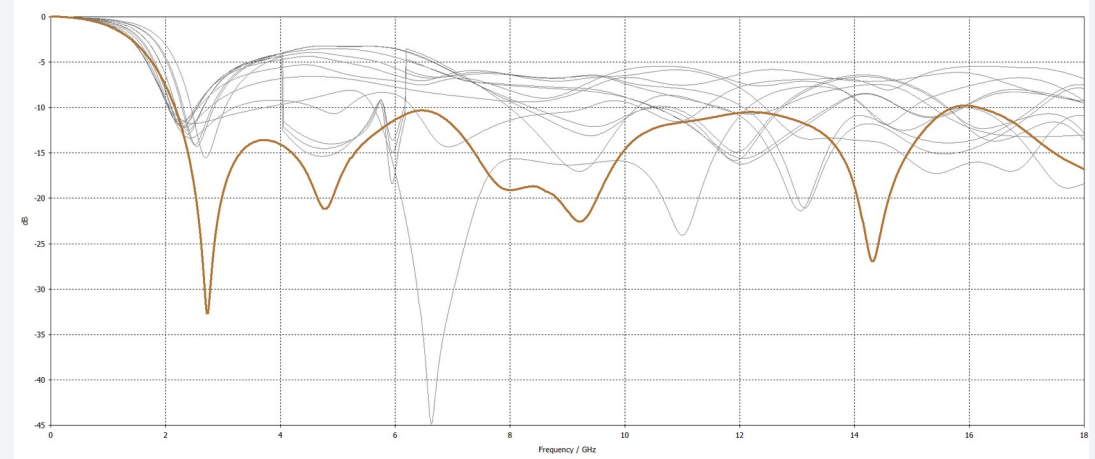
— other values

10 positions: $L_g = -20 \rightarrow -7$ mm

Closer ground $\rightarrow$ stronger coupling $\rightarrow$ better low-band match



Step 2 • Patch size



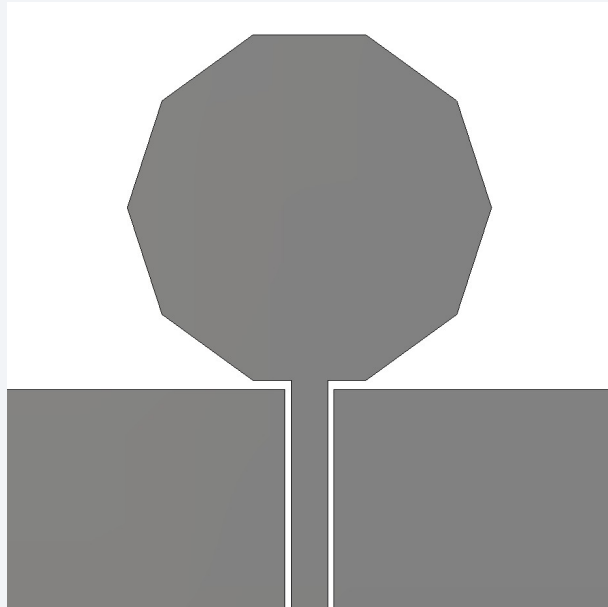
— Best: $R = 15$ mm

— other values

13 sizes: $R = 4 \rightarrow 15$ mm

Bigger patch $\rightarrow$ resonances join into one band

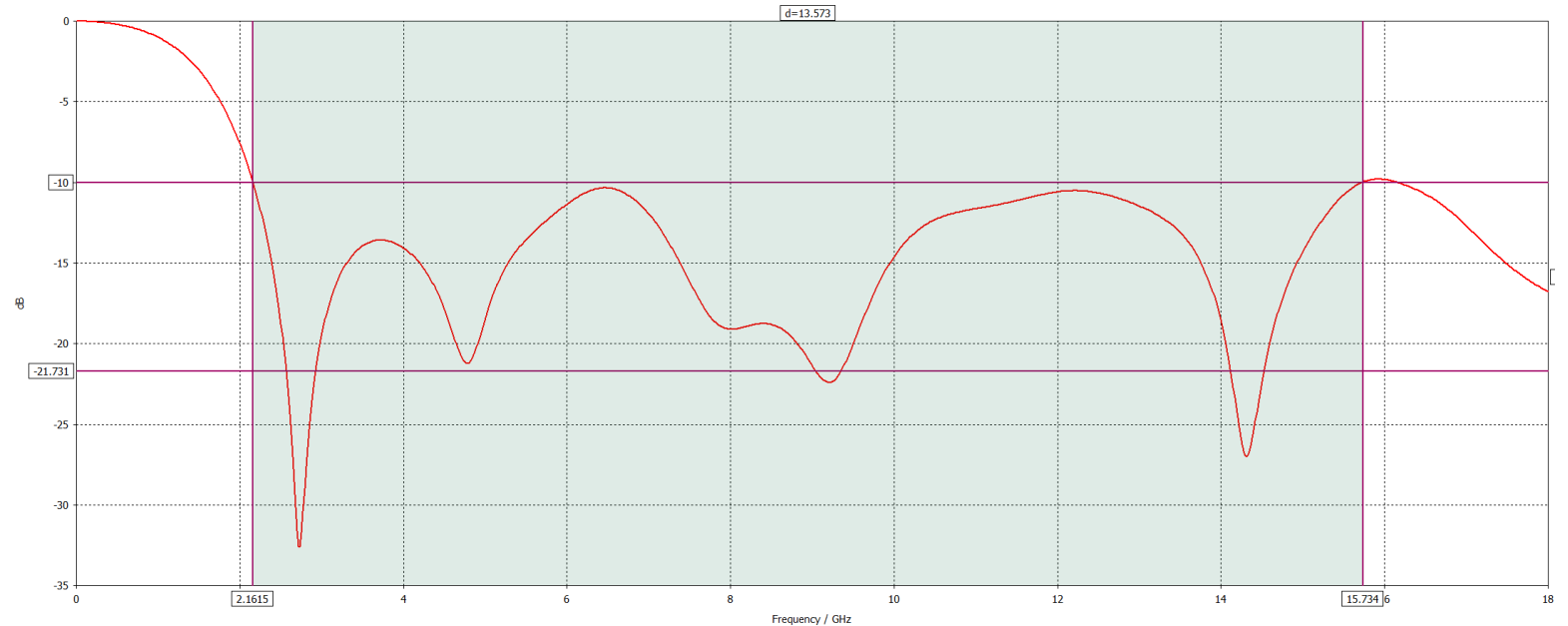
Optimised antenna: matched from 2.16 to 15.73 GHz



Board FR-4, $50 \times 50 \times 1.6$ mm

Patch decagon, $R = 15$ mm

Ground edge $L_g = -7$ mm



2.16 – 15.73 GHz

S_{11} below -10 dB (green band)



about 13.6 GHz wide



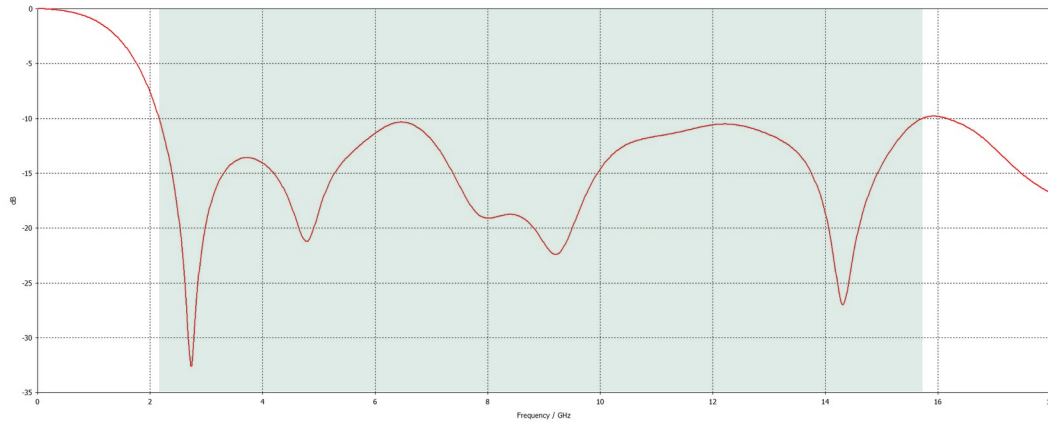
resonances: 2.7 · 4.8 · 9.2 · 14.3 GHz



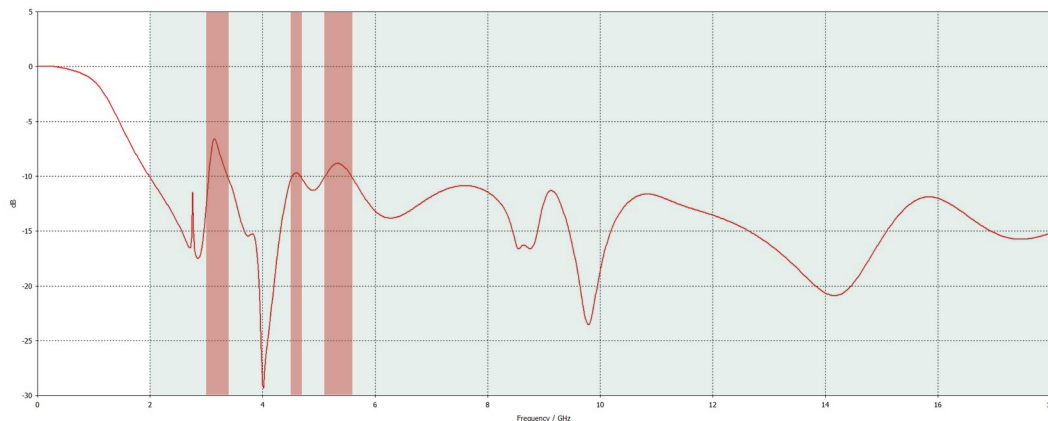
covers the 2–15 GHz target

Metasurface added: wider band, three small gaps

Antenna alone



Antenna + metasurface (2 mm gap)



Wider band

about 2.0 → 18 GHz



3 narrow gaps

3.0–3.4 • 4.5–4.7 • 5.1–5.6 GHz
(worst –6.6 dB near 3.1 GHz)

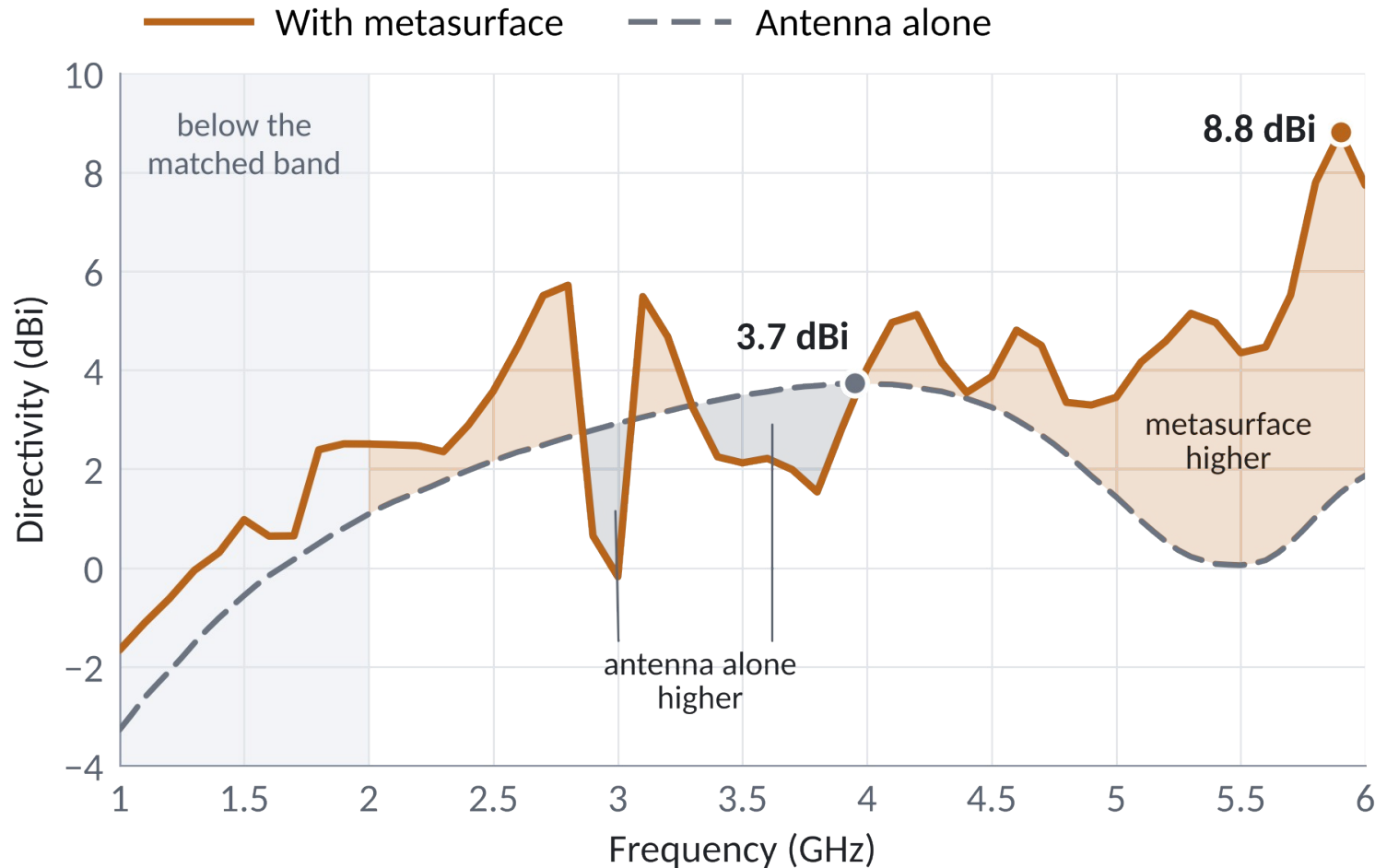


Next: tune

air gap, ring size and ground position

18 GHz is the end of the simulated range • gain: next slide

Metasurface raises the directivity: 3.7 → 8.8 dBi



Peak 3.7 → 8.8 dBi

+5.1 dB with the metasurface
(peak now at 5.9 GHz)



Higher on 78 % of 2–6 GHz

average 2.3 → 3.9 dBi (+1.6 dB)



Two dips

≈ 2.9–3.1 and 3.3–4.0 GHz
(worst –3.1 dB at 3.0 GHz)

CST directivity, maximum in the $\phi = 0^\circ$ plane

Metasurface run covers 1–6 GHz

Directivity = gain without losses; realized gain is next

What were the challenges, and what comes next

Challenges



Long run times

antenna + metasurface needs a large mesh



Sweep glitches

jumps near 4 and 6 GHz from too few frequency points
→ re-run



Thin margin

only just below -10 dB near 6.5 and 12.2 GHz



Metasurface detuning

three narrow S_{11} gaps (3.0–5.6 GHz) and two directivity dips near 3–4 GHz

Remaining in this phase

- 1 Realized gain over the whole 2–15 GHz band
- 2 Tune the metasurface: close the S_{11} gaps and the directivity dips
- 3 Radiation patterns at a few frequencies

Phase II

- 4 4-port MIMO (elements turned 90°) + metasurface
- 5 Isolation and correlation (ECC) between ports
- 6 Fabricate on FR-4 and measure

Expected outcomes

Thin wideband antenna with higher gain

Compact MIMO antenna, good isolation

Tested prototype vs simulation

Conclusion



Antenna done

CPW-fed decagon matched from
2.16–15.73 GHz



Metasurface added

6×5 split rings, 2 mm behind: wider
band, **peak directivity 3.7 → 8.8 dBi**



Next

Realized gain and tuning
→ MIMO antenna

References

- [1] N. P. Agrawal, G. Kumar, and K. P. Ray, "Wide-band planar monopole antennas," IEEE Trans. Antennas Propag., vol. 46, no. 2, pp. 294–295, Feb. 1998.
- [2] M. S. Sharawi, "Printed multi-band MIMO antenna systems and their performance metrics," IEEE Antennas Propag. Mag., vol. 55, no. 5, pp. 218–232, Oct. 2013.
- [3] C. A. Balanis, Antenna Theory: Analysis and Design, 4th ed. Hoboken, NJ, USA: Wiley, 2016.
- [4] R. N. Simons, Coplanar Waveguide Circuits, Components, and Systems. New York, NY, USA: Wiley, 2001.
- [5] C. L. Holloway et al., "An overview of the theory and applications of metasurfaces," IEEE Antennas Propag. Mag., vol. 54, no. 2, pp. 10–35, Apr. 2012.
- [6] J. B. Pendry, A. J. Holden, D. J. Robbins, and W. J. Stewart, "Magnetism from conductors and enhanced nonlinear phenomena," IEEE Trans. Microw. Theory Techn., vol. 47, no. 11, pp. 2075–2084, Nov. 1999.
- [7] A. J. A. Al-Gburi et al., "High gain of UWB planar antenna utilising FSS reflector for UWB applications," Comput. Mater. Contin., vol. 70, no. 1, pp. 1425–1436, 2022.
- [8] M. Hussain et al., "Bandwidth and gain enhancement of a CPW antenna using frequency selective surface for UWB applications," Micromachines, vol. 14, no. 3, Art. no. 591, 2023.
- [9] B. Hammache et al., "Gain enhancement of compact CPW-fed ultra-wideband antenna using an FSS reflector," Microw. Opt. Technol. Lett., vol. 66, no. 10, Art. no. e34344, 2024.
- [10] G. Sen, A. Banerjee, M. Kumar, and S. Das, "An ultra-wideband monopole antenna with a gain enhanced performance using a novel split-ring meta-surface reflector," Microw. Opt. Technol. Lett., vol. 59, no. 6, pp. 1296–1300, 2017.
- [11] M. M. Hasan et al., "Gain and isolation enhancement of a wideband MIMO antenna using metasurface for 5G sub-6 GHz communication systems," Sci. Rep., vol. 12, Art. no. 9433, 2022.